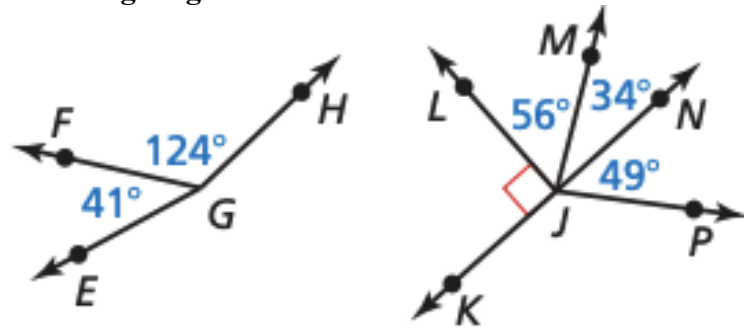


Assignment 1.9b: Describing Angle Pairs

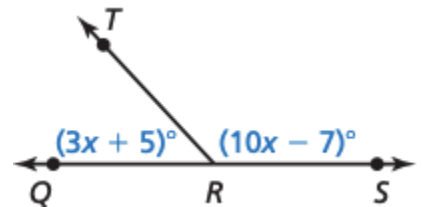
1. Name a pair of the indicated angles.
 - a. Adjacent Complementary Angles
 - b. Adjacent Supplementary Angles
 - c. Nonadjacent Complementary Angles
 - d. Nonadjacent Supplementary Angles



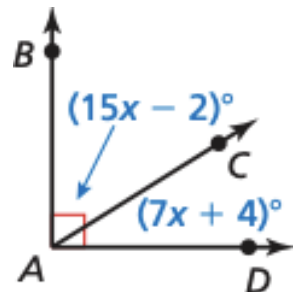
Use the given information to find the measure of the unknown angle.

2. $\angle 3$ is a complement of $\angle 4$, and $m\angle 3 = 46^\circ$
3. $\angle 7$ is a supplement of $\angle 8$, and $m\angle 7 = 109^\circ$

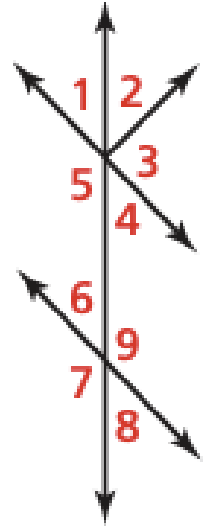
4. Find the measure of each angle.
 - a. Set up an equation that represents the relationship in the diagram.
 - b. Solve for x
 - c. Find $m\angle QRT$ and $m\angle TRS$



5. Find the measure of each angle.
 - a. Set up an equation that represents the relationship in the diagram.
 - b. Solve for x
 - c. Find $m\angle BAC$ and $m\angle CAD$



6. Use the diagram to answer the following questions.
- a. Identify all the linear pairs that include $\angle 5$.



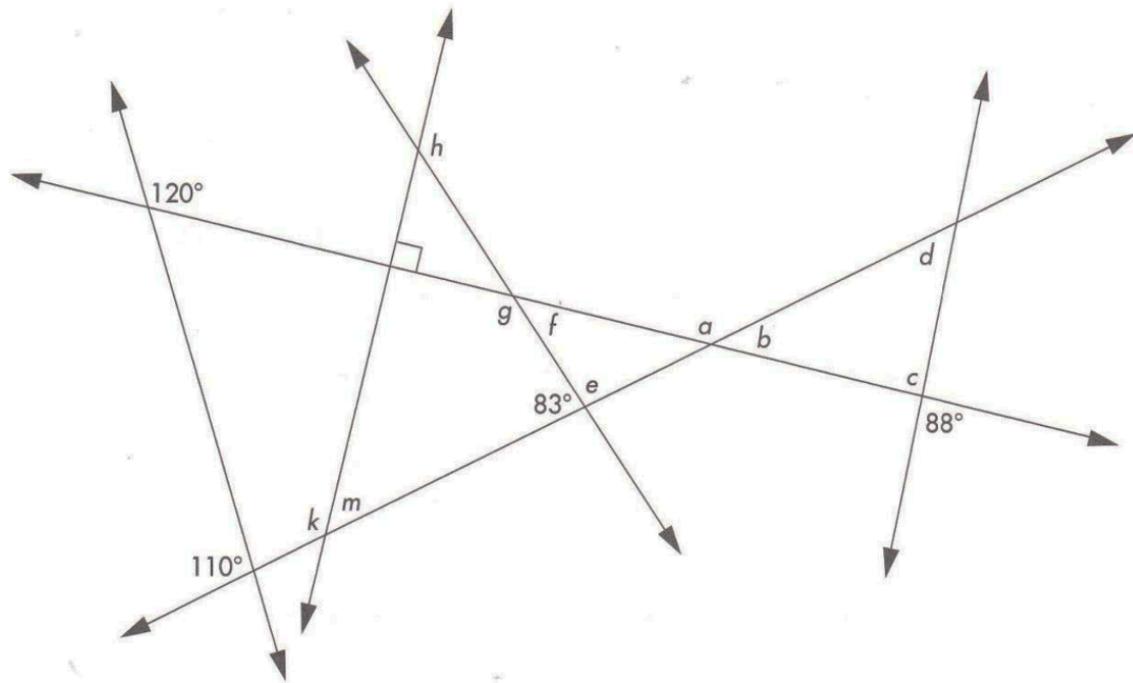
- b. Identify all the linear pairs that include $\angle 6$.

- c. Are $\angle 6$ and $\angle 8$ vertical angles?

- d. Are $\angle 2$ and $\angle 5$ vertical angles?

7. Two angles form a linear pair. The measure of the smaller angle is 102° less than the measure of the larger angle. Find the measure of each angle.
8. Two angles are complementary. The measure of the larger angle is 5 times the measure of the smaller angle. Find the measure of each angle.
9. Two angles are supplementary. The measure of the larger angle is 6° less than twice the measure of the smaller angle. Find the measure of each angle.

10. Fun fact: The angles inside of a triangle must add to 180° . Use this fact to find all of the missing angles in the image below. Show your work on the paper below when possible.



$a =$

$b =$

$c =$

$d =$

$e =$

$f =$

$g =$

$h =$

Optional Challenge

Two lines intersect at point O , forming four angles. The measure of one angle is three times the measure of an adjacent angle.

- Draw a picture of this scenario.
- Find the measures of all four angles. Explain how you know.
- A ray is drawn from O , dividing the smaller angle into two complementary angles. Is this possible? Explain.
- A ray is drawn from O , dividing the larger angle into two angles, one of which is twice the other. Find the measures of the two new angles.
- Create a different pair of intersecting lines where one angle is four times its adjacent angle. Find all four angle measures.